

Introduction to OSINT

Open Source Intelligence (OSINT) refers to the process of collecting, analyzing, and utilizing information that is publicly available from various sources such as websites, social media platforms, public records, search engines, DNS records, and online databases. OSINT is widely used in cybersecurity, digital forensics, threat intelligence, law enforcement, and security assessments to gather information about individuals, organizations, domains, and network infrastructure.

The primary objective of OSINT is to transform publicly accessible data into meaningful intelligence that can support investigations, risk assessments, and security analysis. Since the information is collected from legal and publicly available sources, OSINT provides a cost-effective and efficient method for understanding an organization's digital footprint and identifying potential security exposures.

In cybersecurity, OSINT plays a crucial role during the reconnaissance phase, where analysts gather information about a target before conducting further security assessments. Common OSINT tools include Maltego, Shodan, Recon-ng, SpiderFoot, and theHarvester.

Introduction to Maltego

Maltego is a powerful Open Source Intelligence (OSINT) and link analysis tool developed by Paterva. It is widely used by cybersecurity professionals, investigators, researchers, and intelligence analysts to discover relationships between entities such as domains, IP addresses, websites, organizations, email addresses, social media accounts, and individuals.

Maltego uses a graphical interface to visualize connections between different pieces of information, making it easier to understand complex relationships and identify patterns. Through the use of automated queries known as **Transforms**, Maltego collects data from various public sources and presents it in the form of an interactive graph.

The tool is commonly used for domain investigations, network reconnaissance, digital footprint analysis, threat intelligence gathering, fraud investigations, and cybersecurity research. By mapping relationships between entities, Maltego helps analysts gain valuable insights into an organization's online presence and publicly exposed information.

In this experiment, Maltego Community Edition was used to investigate the domain **openai.com**, discover associated subdomains, identify publicly available information, and visualize the relationships between various entities through graph-based analysis.

PART 1: INSTALLATION

1. **Install Maltego Community Edition.**
2. **Create an account and log in.**
3. **Explore the following sections:**
 - **Graph**
 - **Entity Palette**
 - **Transform Hub**
 - **Property View**

Aim

To install Maltego Community Edition, create an account, log in, and explore the basic components of the Maltego interface.

Installation Procedure

Step 1: Download Maltego

1. Open a web browser.
2. Visit the official Maltego website:
[Maltego Official Website](#)
3. Navigate to the Downloads section.
4. Download the appropriate installer for your operating system (Windows/Linux/macOS).

Step 2: Install Maltego

1. Run the downloaded installer.
2. Accept the license agreement.
3. Choose the installation location.
4. Click **Install**.
5. Wait for the installation to complete.
6. Click **Finish**.

Step 3: Create a Maltego Account

1. Launch Maltego.
2. Click **Register** or **Create Account**.
3. Enter:
 - Name
 - Email Address
 - Password
4. Verify the email address if required.
5. Complete the registration process.

Step 4: Log In

1. Open Maltego.
2. Enter your registered email address and password.
3. Click **Login**.
4. Wait for the Maltego dashboard to load.

Exploring the Maltego Interface

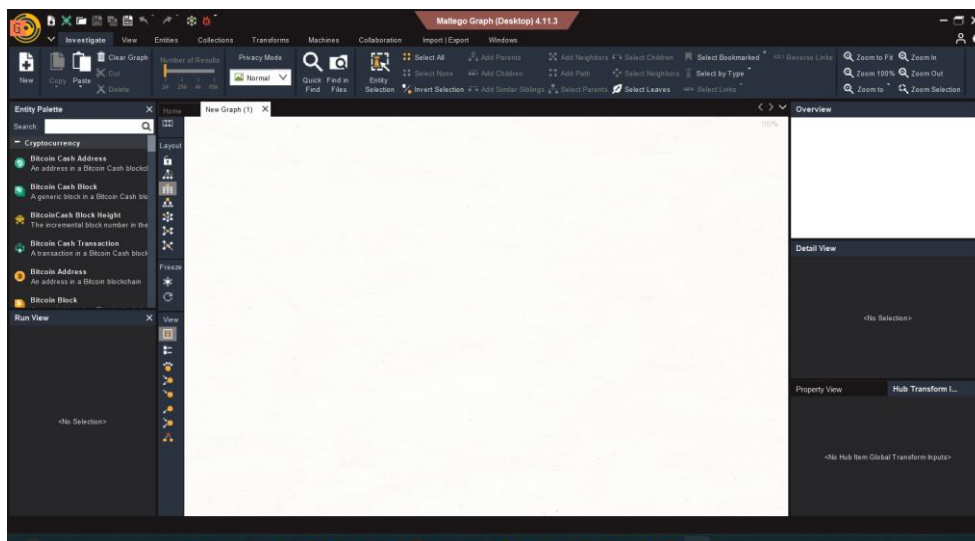
1. Graph Window

The Graph Window is the main workspace where entities and their relationships are displayed visually.

Purpose:

1. Displays investigation results.
2. Shows connections between entities.
3. Allows graph analysis and visualization.

Screenshot



2. Entity Palette

The Entity Palette contains different entity types that can be dragged onto the graph.

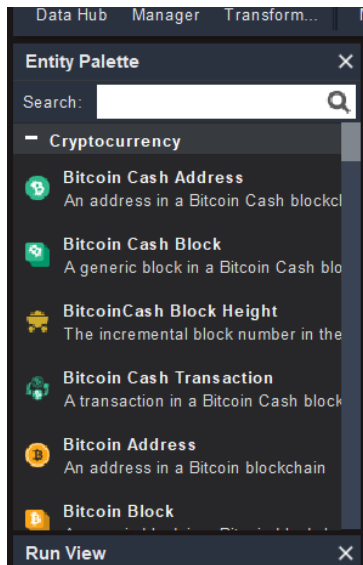
Examples:

1. Domain
2. Website
3. IP Address
4. Person
5. Company
6. Email Address

Purpose:

1. Add entities to investigations.
2. Define starting points for OSINT analysis.

Screenshot



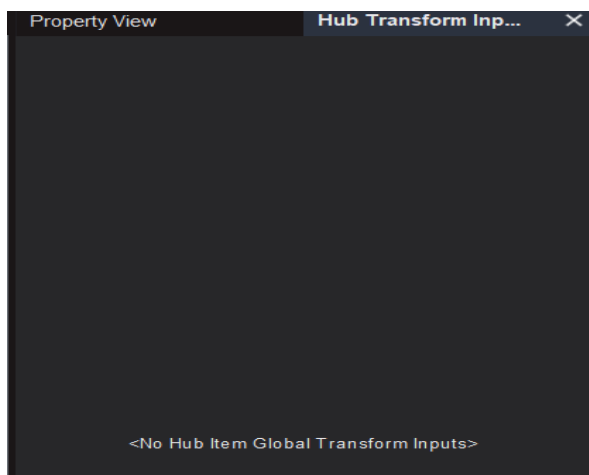
3. Transform Hub

The Transform Hub provides transforms used to gather information from various public sources.

Purpose:

1. Discover domains
2. Find IP addresses
3. Retrieve DNS information
4. Collect WHOIS data
5. Perform OSINT investigations

Screenshot



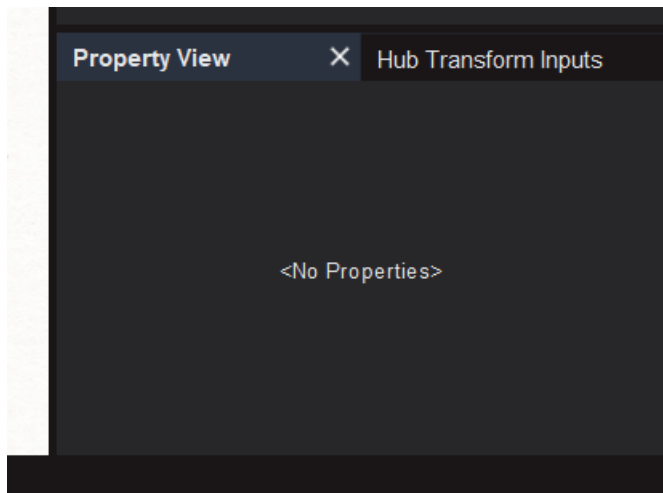
4. Property View

The Property View displays detailed information about the selected entity.

Purpose:

1. View entity properties
2. Edit entity values
3. Review collected information

Screenshot



PART 2: DOMAIN INVESTIGATION

Choose any company or organization.

Examples:

- Google
- Microsoft
- Amazon
- OpenAI

Tasks:

1. Add a Domain Entity.
2. Run available transforms.
3. Collect information about:
 - Domain Names
 - Subdomains
 - IP Addresses
 - Email Addresses
 - Related Websites

Take screenshots of the generated graph.

STEP 1: Add a Domain Entity

Procedure

1. Open **Maltego**.
2. Click **New Graph**.
3. In the **Entity Palette** (left side), search for:
Domain
4. Drag the **Domain** entity onto the blank graph workspace.
5. Double-click the Domain entity.
6. Enter the target domain:
openai.com
7. Press **Enter**.

Screenshot :

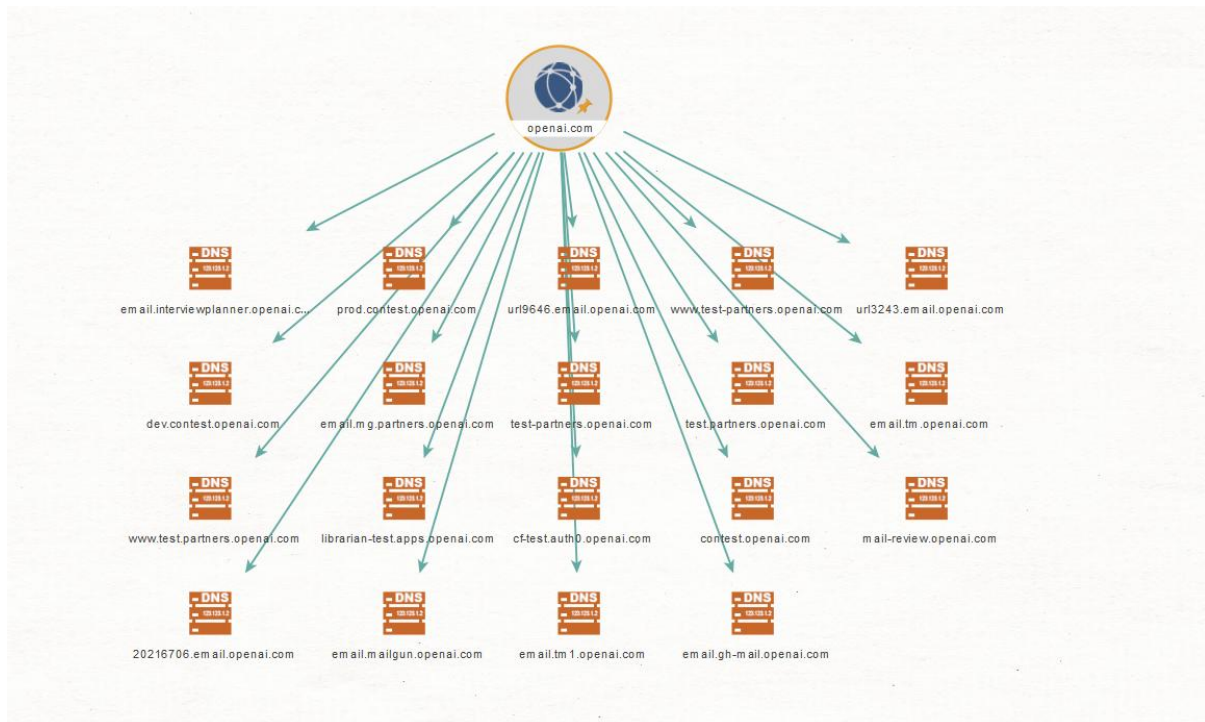


STEP 2: Run DNS Name Transform (Discover Subdomains)

Procedure

1. Right-click on the **openai.com** Domain Entity.
2. Select **Run Transform**.
3. Choose a DNS-related transform such as:
To DNS Name
or
[Utilities] To DNS Name (interesting)
4. Wait for the transform to complete.

Screenshot :

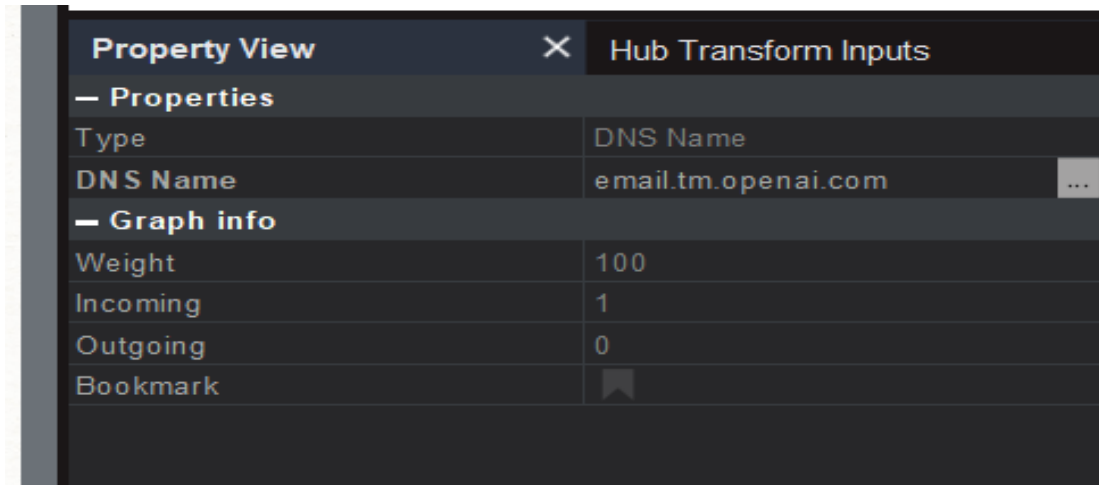


STEP 3: Entity Property Analysis

Procedure

1. Click on one of the discovered subdomains (for example, **admin.openai.com** or any subdomain generated in your graph).
2. Open the **Property View** panel (usually on the right side of Maltego).
3. Observe the information displayed for the selected entity.

Screenshot :

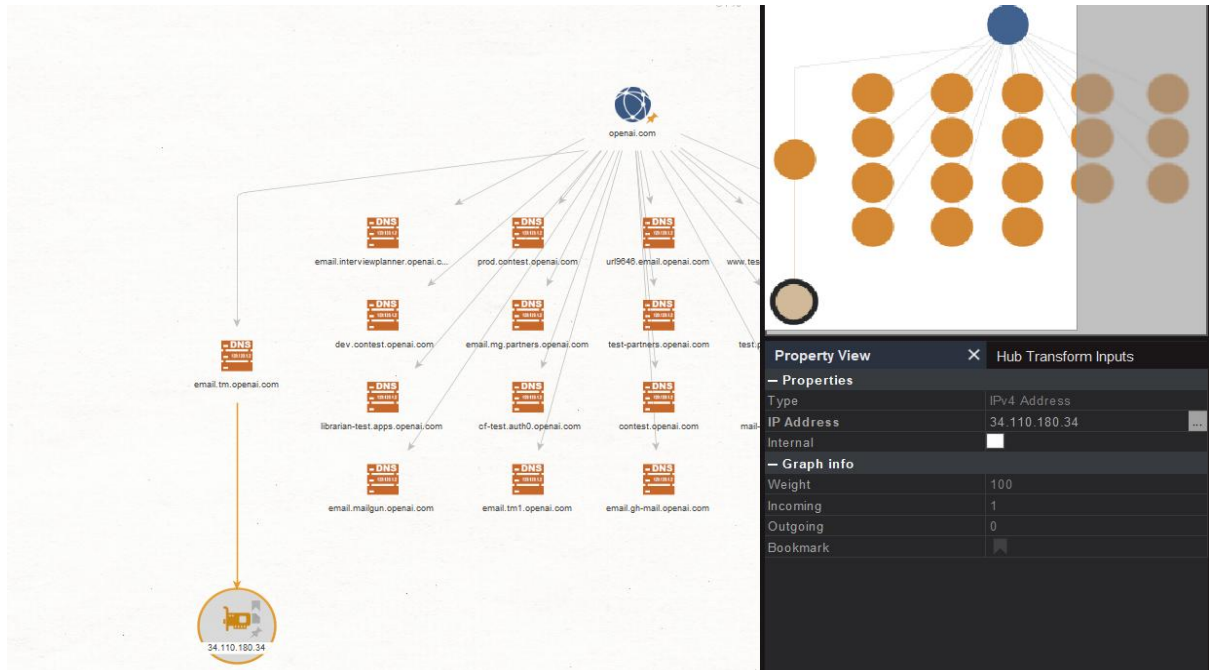


STEP 4: Discover IP Addresses

Procedure

1. Select any discovered subdomain from the graph.
2. Example:
contest.openai.com
3. Right-click the selected DNS Name entity.
4. Click **Run Transforms**.
5. Search for and run a transform such as:
To IP Address
6. Wait for the transform to complete.

Screenshot :

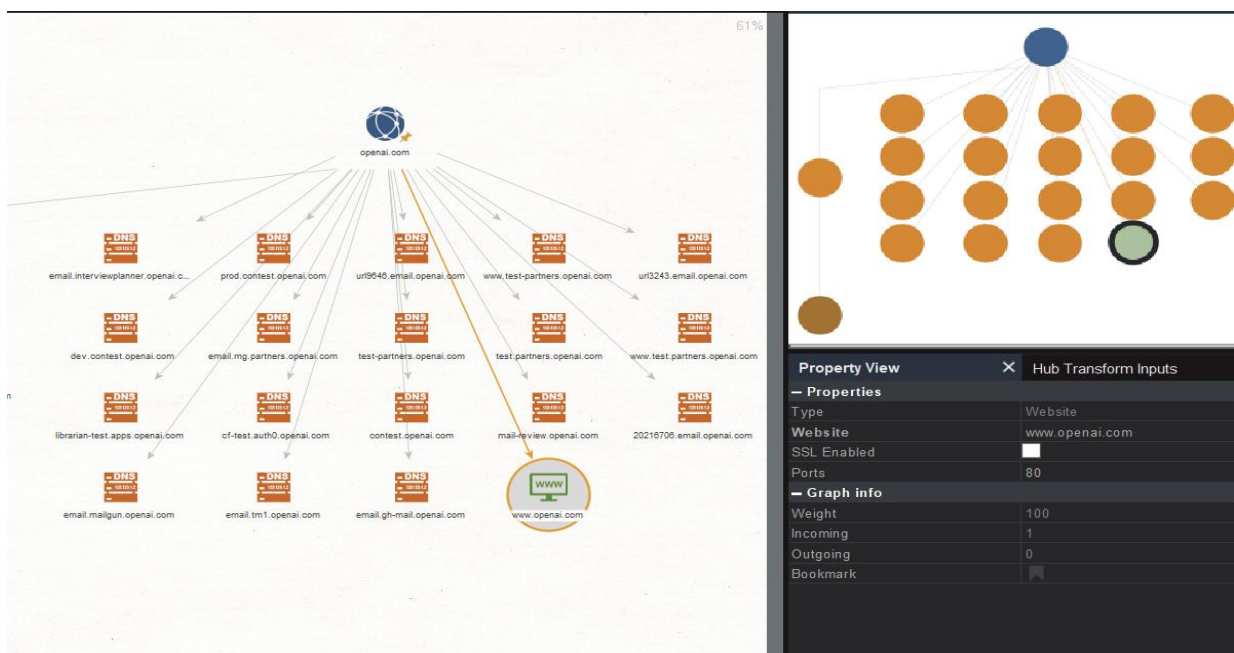


STEP 5: Discover Related Websites and WHOIS Information

Procedure A: Find Related Websites

1. Right-click on the **openai.com** Domain Entity.
2. Click **Run Transforms**.
3. Search for and run: To Website

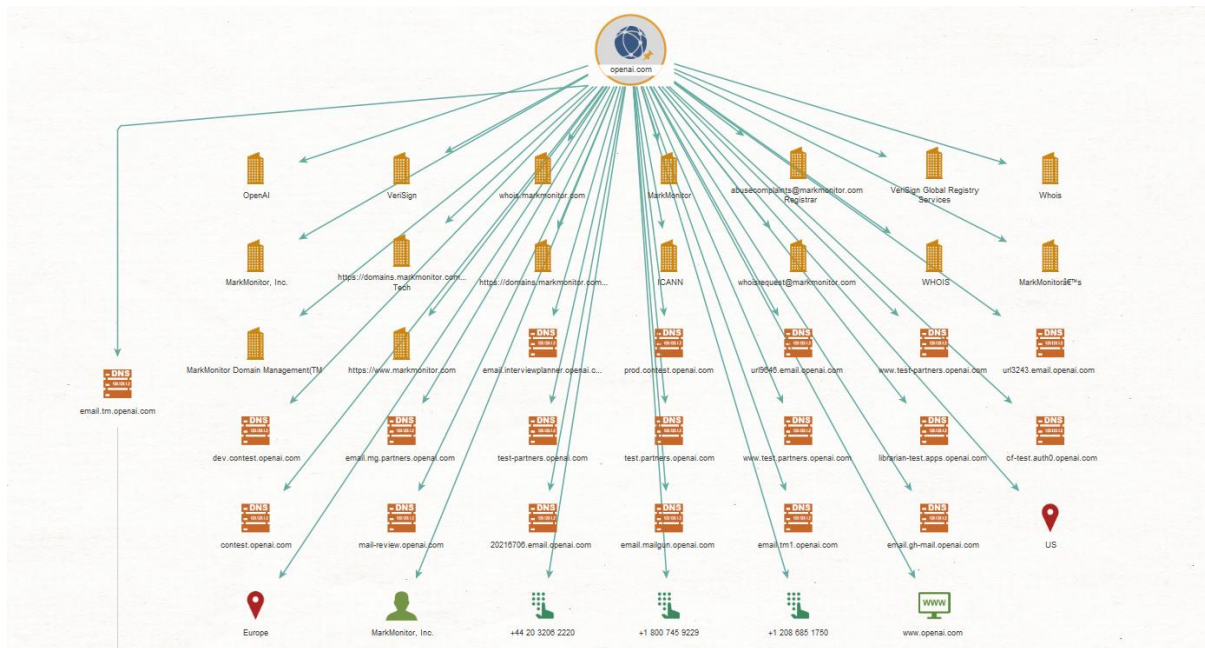
Screenshot :



Procedure B: Collect WHOIS Information

1. Right-click **openai.com**.
2. Search for transforms such as:
WHOIS Information
3. Run the available WHOIS transform.

Screenshot :



Information Collected So Far

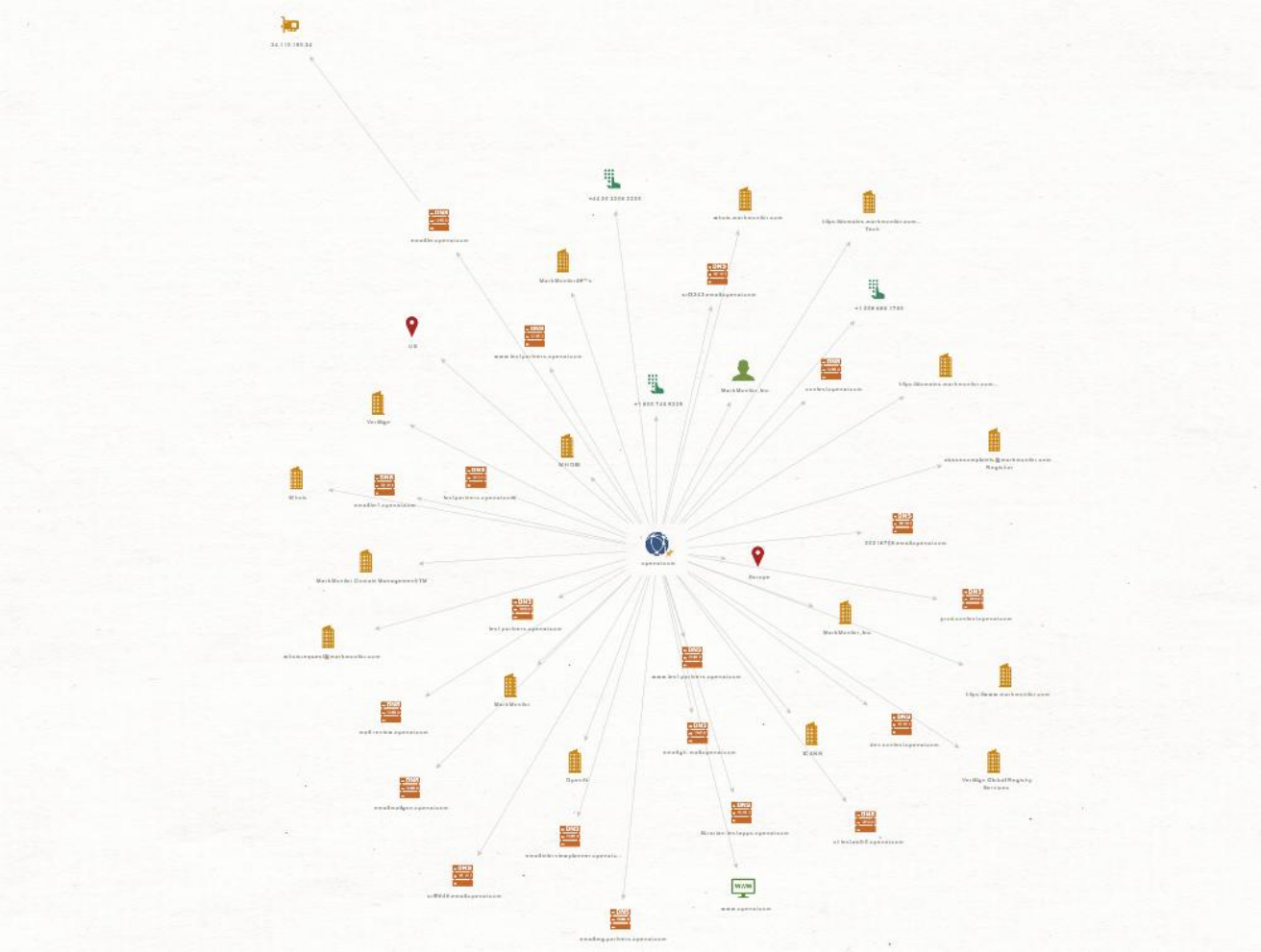
Information Type	Information Collected
Domain Name	openai.com
Subdomains	email.tm.openai.com, dev.contest.openai.com, contest.openai.com, mail-review.openai.com, email.interviewplanner.openai.com, prod.contest.openai.com, test-partners.openai.com, www.test-partners.openai.com , librarian-test.apps.openai.com, cf-test.auth0.openai.com, email.mailgun.openai.com, email.tml.openai.com
IP Addresses	IP addresses discovered through the IP Address transforms (insert the IPs generated in your graph if available)
Email Addresses / Contact Information	whoisrequest@markmonitor.com
Phone Numbers	+44 20 3206 2220, +1 800 745 9229, +1 208 685 1750
Related Websites	www.openai.com , www.markmonitor.com , domains.markmonitor.com
Registrar Information	MarkMonitor Inc., MarkMonitor Domain Management™, VeriSign, ICANN
Organization	OpenAI
Location Information	US, Europe

STEP 6: Final Graph Analysis and Documentation

Procedure

- Ensure all discovered entities are visible in the graph:
 - Domain (openai.com)
 - Subdomains
 - Related Websites
 - Registrar Information
 - Contact Information
 - Locations
 - Phone Numbers
- Arrange the graph using: **Layout** → **Organic Layout**
- Adjust the graph so that all relationships are clearly visible.
- Capture the final graph screenshot.

Screenshot :



Information Collected :

Information Type	Information Collected
Domain Name	openai.com
Organization	OpenAI
Subdomains	Multiple subdomains discovered
Related Website	www.openai.com
Registrar	MarkMonitor Inc.
Contact Email	whoisrequest@markmonitor.com
Phone Numbers	+44 20 3206 2220, +1 800 745 9229, +1 208 685 1750
Locations	US, Europe

PART 3: EMAIL & COMPANY RESEARCH

Find and document:

- 1. Official Website**
- 2. Company Headquarters**
- 3. Social Media Accounts**
- 4. Public Contact Information**
- 5. Technologies used by the organization (if available)**

Official Website

OpenAI (www.openai.com)

OpenAI's official website serves as the primary online platform for the organization. It provides information about OpenAI's research initiatives, artificial intelligence technologies, products, services, and company announcements. The website also offers access to ChatGPT, API services, developer resources, safety research publications, and documentation for AI models.

Headquarters

OpenAI Headquarters – San Francisco, California, United States

OpenAI is headquartered in San Francisco, California, USA. The headquarters serves as the central location for the company's research, development, business operations, and strategic decision-making. From this location, OpenAI manages its AI research projects, product development, partnerships, and global operations.

Social Media Accounts

• OpenAI X (Twitter)

OpenAI uses X (formerly Twitter) to share company announcements, research updates, product releases, safety initiatives, and AI-related news.

• OpenAI LinkedIn

LinkedIn is used for professional networking, recruitment, company updates, and sharing information about OpenAI's achievements and career opportunities.

• OpenAI YouTube

The YouTube channel contains product demonstrations, conference presentations, educational content, interviews, and updates related to artificial intelligence research.

• OpenAI GitHub

GitHub hosts open-source projects, research code, technical resources, and collaborative development initiatives associated with OpenAI.

Public Contact Information

- **WHOIS Contact Details**

WHOIS records provide publicly accessible information related to the registration and management of the OpenAI domain.

- **Public Support Information**

OpenAI offers support services through official help centers, documentation portals, and customer support channels available on its website.

- **Contact Email**

- whoisrequest@markmonitor.com

- **Public Phone Numbers**

- +44 20 3206 2220
- +1 800 745 9229
- +1 208 685 1750

- **Domain Registrar**

- MarkMonitor Inc.

These contact details are publicly available through domain registration and management records and are used for administrative and domain-related communications.

Technologies Used

- **Cloudflare**

Cloudflare provides security, content delivery, and performance optimization services. It helps protect websites from cyberattacks and improves website availability and speed.

- **Web Hosting Infrastructure**

OpenAI utilizes modern hosting infrastructure to support its websites, APIs, and online services, ensuring scalability and reliability.

- **DNS Services**

Domain Name System (DNS) services are used to translate domain names into IP addresses, allowing users to access OpenAI's online resources efficiently.

- **Security Certificates (SSL/TLS)**

SSL/TLS certificates encrypt communication between users and OpenAI's services, ensuring secure data transmission and protecting sensitive information.

- **Domain Management Services (MarkMonitor)**

MarkMonitor is responsible for managing and protecting OpenAI's domain registrations and digital assets.

- **VeriSign Registry Services**

VeriSign provides domain registry infrastructure and services that support domain name operations and internet stability.

- **Email Infrastructure**

Email-related technologies and mail servers support communication, notifications, and operational services associated with OpenAI's domain.

- **Cloud-Based Services**

Cloud infrastructure enables OpenAI to deliver scalable AI applications, APIs, and online services to users worldwide.

PART 4: VISUAL ANALYSIS

Create a relationship diagram showing:

Company



Website



Subdomains



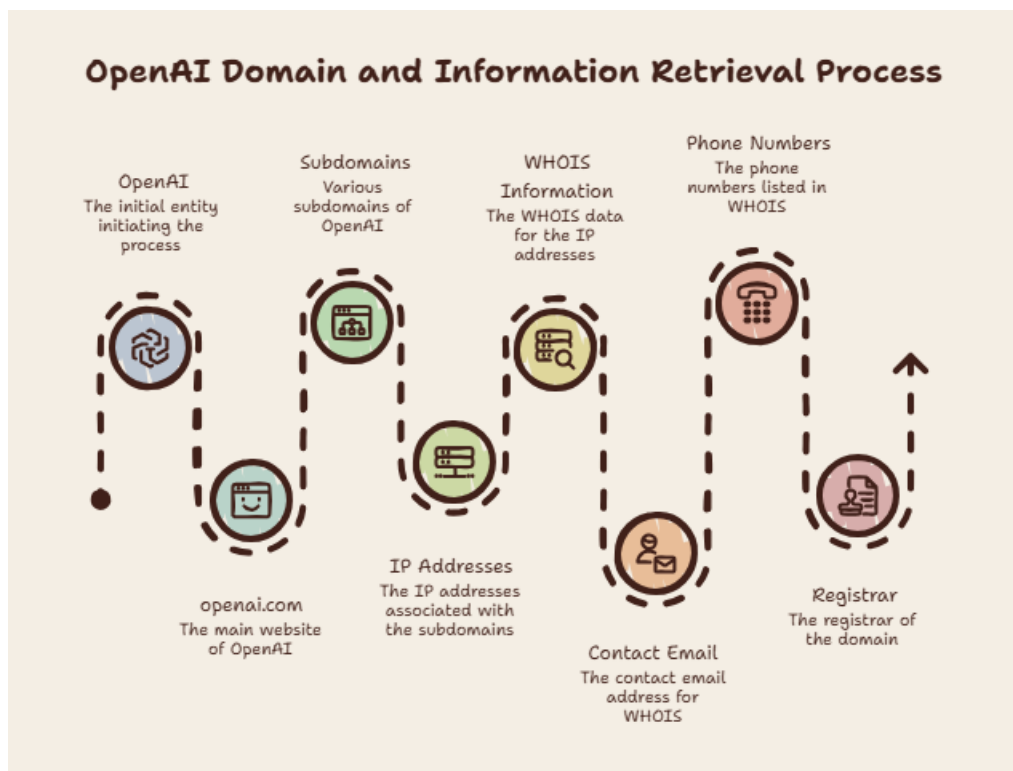
IP Addresses



Public Information

Explain the relationships discovered.

Relationship Diagram



Relationship Analysis

The investigation began with the organization **OpenAI** and its primary domain **openai.com**. Using Maltego transforms, several subdomains associated with the domain were discovered, including **contest.openai.com**, **email.tm.openai.com**, **test-partners.openai.com**, and other related services.

These subdomains were linked to various public IP addresses, revealing the infrastructure used to host and manage OpenAI's online services. Further analysis through WHOIS records

provided publicly available registration information, including contact email addresses, phone numbers, registrar details, and geographic information.

The relationship graph demonstrates how different components of an organization's online presence are interconnected. Starting from a single domain, it is possible to discover subdomains, network infrastructure, contact information, and domain registration details. This highlights the effectiveness of OSINT tools in mapping an organization's digital footprint and understanding publicly exposed information.

Security Risks Identified

1. Exposure of Subdomains

Publicly accessible subdomains may reveal development, testing, or administrative services.

Risk: Attackers may target these systems for exploitation.

2. Exposure of Contact Information

Email addresses and phone numbers may be publicly available.

Risk: Can lead to phishing, spam, and social engineering attacks.

3. Exposure of Infrastructure Information

IP addresses and DNS records reveal network details.

Risk: Enables targeted scanning and vulnerability assessments.

4. Exposure of Registrar Information

Domain registration details may be publicly accessible.

Risk: May facilitate domain hijacking or impersonation attacks.

5. Information Aggregation

Small pieces of public information can be combined.

Risk: Attackers can build detailed profiles of the organization.

Recommendations for Organizations

1. Limit exposure of internal and testing systems.
2. Use strong authentication and Multi-Factor Authentication (MFA).
3. Regularly review and update DNS records.
4. Protect sensitive contact information.
5. Conduct periodic security assessments.
6. Monitor publicly available information using OSINT tools.
7. Secure domain management with registrar locks.
8. Keep systems and security configurations updated.

ASSIGNMENT QUESTIONS

1. How many domains are associated with the company?

Answer:

The investigation identified **1 primary domain (openai.com)** and **more than 10 associated subdomains**, including contest.openai.com, dev.contest.openai.com, email.tm.openai.com, test-partners.openai.com, and others.

2. What public information is exposed?

Answer:

1. Domain name information
2. Subdomains
3. Website URLs
4. WHOIS records
5. Contact email addresses
6. Public phone numbers
7. Registrar details
8. Geographic information
9. DNS-related information

3. What sensitive information should organizations avoid exposing publicly?

Answer:

1. Internal administrative portals
2. Employee email addresses
3. Internal IP addresses
4. Authentication endpoints
5. API keys and access tokens
6. Development and testing environments
7. Server configuration details
8. Sensitive business information